UNIWERSYTET VIZJA · SCHOOL OF COMPUTER SCIENCE & TECHNOLOGIES

Design and Implementation of a **Web-Based Student Management System** for Educational Institutions

VIZJA UNIVERSITY

LIVE https://thesms.me

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Diploma seminar

FIELD OF STUDY
Computer Science
Engineering degree · 2026

WHAT THIS DEFENCE COVERS

Presentation Agenda

Problem Statement & Purpose

3 places
spreadsheet, paper register and email thread

0
audit trail of who entered or edited a grade

by hand
every average and every attendance total

THE PROBLEM

- **Fragmented records —** spreadsheets, paper registers, email threads
- **No single view —** grades and attendance never meet in one place
- **Manual arithmetic —** averages recalculated by hand, errors slip through
- **No accountability —** nobody can prove who recorded a mark
- **No confidentiality —** every teacher can see every student
- **No Polish interface —** staff work in English-only tools

THE PURPOSE

- **One source of truth —** a relational schema with enforced constraints
- **Instant analytics —** KPI tiles and three charts from ORM aggregation
- **Automatic authorship —** every grade stores who assigned it
- **Scoped visibility —** teachers see their own courses, admins see all
- **Bilingual by design —** the whole interface switches EN ⇄ PL
- **Deployable —** one build script, HTTPS, managed PostgreSQL

Result: one auditable, role-aware platform now managing 25 students, 8 courses, 87 enrollment records — 51 of them active — and 87 grades in production.

Comparative Analysis

CRITERION	MOODLE	USOS	GOOGLE CLASSROOM	SMS — THIS PROJECT
Target user	Large universities	Polish public HEIs	Schools and courses	**Departments and small institutions**
Deployment	Self-hosted, heavy	Vendor-managed	Google cloud only	**Self-hosted, one Django app**
Setup effort	Days of plugin work	Institutional contract	Minutes, no control	**Minutes — a single build script**
Gradebook	Yes, complex	Yes	Basic	**Weighted components + author audit**
Attendance register	Plugin required	Yes	Not available	**Built in, three statuses**
Role isolation	Configurable, intricate	Yes	Limited	**Enforced at queryset level**
Bilingual EN / PL	Language packs	Polish-first	Yes	**Native, 312 strings**
Analytics	Reports plugin	Institutional reports	Minimal	**Chart.js dashboard + CSV**
Cost of ownership	GPL + hosting	Licensed	Free, data in Google	**Open source, €0 hosting**

Positioning: the established platforms are either too heavy or too closed for a single department; SMS trades breadth for a focused, auditable and fully self-hosted core.

Technology Stack

BACKEND

Runtime · data · authentication

- Python 3.12
- Django 6.0.6
- PostgreSQL 16 (Neon)
- SQLite 3 for development
- Django session authentication
- Custom user model with roles
- Django ORM · 6 migrations
- dj-database-url 2.1

FRONTEND

Templates · charts · theming

- Django Templates · 30 files
- Semantic HTML5
- CSS custom properties · 1 945 lines
- Vanilla JavaScript ES6 · 425 lines
- Chart.js 4 — doughnut and bars
- Turbo 8 page transitions
- Bootstrap Icons
- Light and dark themes

INFRASTRUCTURE

Hosting · operations · CI

- Render web service
- Neon managed PostgreSQL
- Gunicorn 22 (WSGI)
- WhiteNoise 6.6 static files
- HTTPS with HSTS preload
- GitHub Actions CI
- Git version control
- Zero CDN dependencies

No build step, no separate API server. Every asset is vendored locally, so the application runs and demonstrates fully offline.

System Architecture

CLIENT

Browser

HTML · CSS · ES6

Turbo navigation

›

EDGE

Render · TLS 443

HTTPS, HSTS

static via WhiteNoise

›

APPLICATION

Gunicorn + Django 6

URL › View › Template

ORM aggregation

›

DATA

PostgreSQL 16

Neon managed

daily backups

REQUEST PIPELINE — MIDDLEWARE

Security · Session · CSRF · Locale · WhiteNoise · Authentication · Role decorators · Queryset scoping

SUPPORTING SERVICES

i18n EN / PL
Interface language switch

GitHub Actions
Tests and deploy gates

Teachers screen
Admin issues lecturer accounts

CSV export
Role-scoped reporting

/api/dashboard/
JSON metrics endpoint

Database Design

User *extends AbstractUser*

id
username · email
first_name · last_name
role → admin | teacher
password (PBKDF2)
is_active

Student *academic identity*

id
student_number (unique)
first_name · last_name
email
study_program
date_enrolled · is_active

Course *taught unit*

id
course_code (unique)
course_name · description
semester · credits
max_students
teacher_id → User

Enrollment *student ⇄ course*

id
student_id → Student
course_id → Course
enrollment_date
status → active | completed | dropped
unique (student, course)

Grade *weighted component*

id
enrollment_id → Enrollment
kind → coursework | midterm | final | retake
weight 1–100 %
grade_value 2.0 – 5.0
assigned_by → User

Attendance *daily register*

id
enrollment_id → Enrollment
date
status → present | absent | late
notes
unique (enrollment, date)

Integrity where it belongs: unique constraints, foreign keys and cascade rules live in the schema, so no interface path can create a duplicate enrollment or an orphaned grade.

Key Features

Academic Records

CRUD for students, courses and enrollments. Administrators issue lecturer accounts on Teachers. Enrollment is refused once max_students is reached.

Weighted Gradebook

Coursework, midterm, exam and retake — each with a weight. The course mark is their weighted mean. assigned_by records who last saved the component.

Attendance Register

Present, absent and late — one row per enrollment per day, enforced in the schema. Mark Class saves a whole roster in a single request.

Analytics & Export

Four KPI tiles, three Chart.js views and JSON at /api/dashboard/. CSV export of students and grades is scoped to the caller's role.

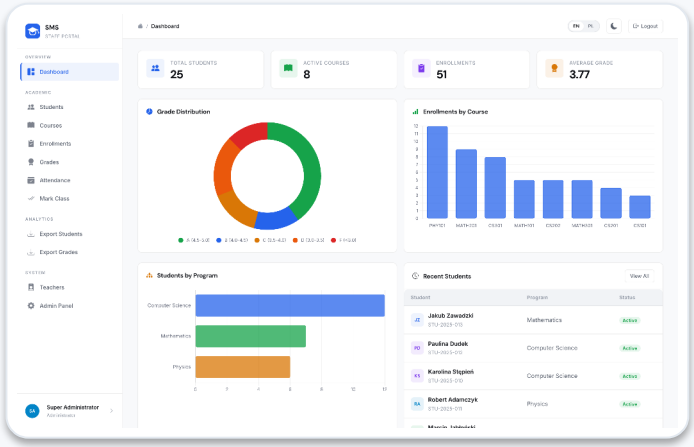
Role-Based Access

Administrators see the institution; teachers see only their courses. Decorators guard the route, querysets filter again — a guessed URL returns nothing.

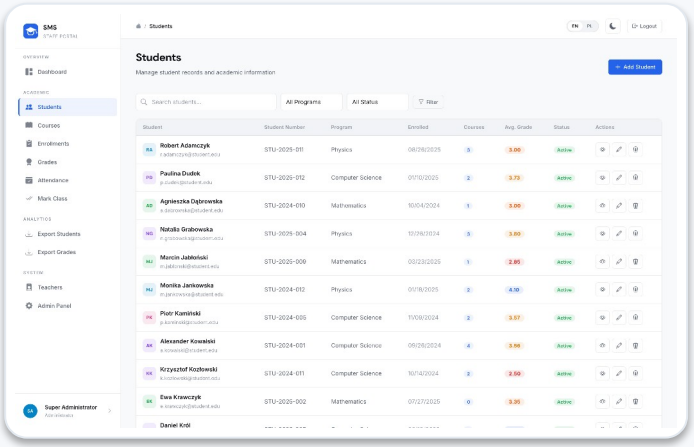
i18n & Quality

312 strings, English ⇄ Polish, locale-aware dates. 96 automated tests and four CI gates (tests, migrations, deploy check, collectstatic) on every push.

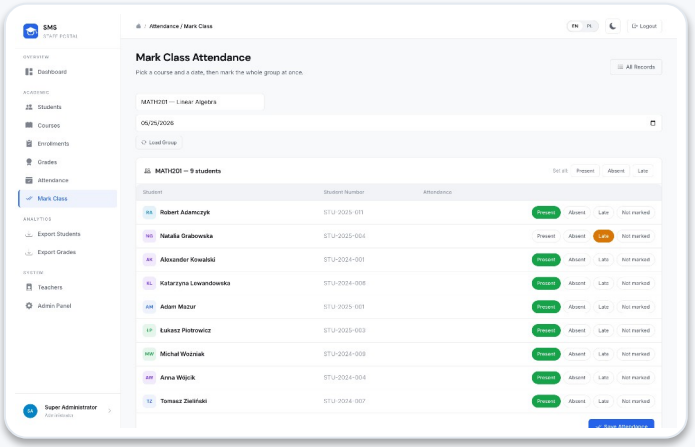
Application Screenshots



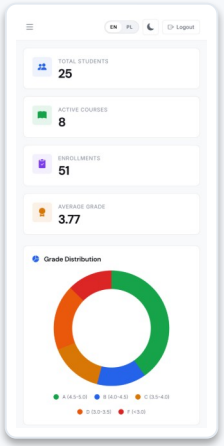
Dashboard — KPI tiles and three Chart.js views



Student register — search, filters, pagination



Mark Class — a whole roster saved in one request



Responsive down to 375 px

Every table is role-aware: a teacher opening the same page sees only students in their own courses, and the action buttons disappear.

IN PRODUCTION, NOT ON A LAPTOP

Deployment & Results

LIVE https://thesms.me

DEFENCE LOGIN (teacher)
prof.martinez / demo1234

25	8	87	87	€0
students	courses	enrollment records	grades	monthly cost

DELIVERED

- **Live on Render + Neon —** HTTPS, HSTS, managed PostgreSQL
- **Six-model schema —** constraints and an audit trail in the database
- **Role isolation —** admins and teachers see strictly their own data
- **Bilingual interface —** 312 strings, locale-aware dates and numbers
- **96 tests, four CI gates —** green on every push, then build.sh deploys

FUTURE WORK

- **Student portal —** a third role with read-only access to own results
- **Semester transcripts —** generated PDF reports per student
- **Email notifications —** alerts on new grades and absences
- **Documented REST API —** token-authenticated public interface
- **Two-factor login —** TOTP for administrator accounts

Thank you for your attention

Questions are welcome.

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